

AYMANE AIT DADS

AI Algorithm Research Intern - Neuromorphic Computing | PyTorch · Transformer Fine-tuning · Unsupervised Learning

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Hiring Team - AI Algorithm Research Intern - Neuromorphic Computing

Intel Corporation · Hillsboro, OR, USA

Dear Intel Neuromorphic Computing Lab Hiring Team,

Intel's Neuromorphic Computing Lab is looking for researchers who can develop **neuro-inspired algorithms** that exploit the **event-driven, sparse processing** paradigm pioneered by the Loihi chips - a constraint that maps directly to my work building a **transformer autoencoder for unsupervised fault detection** on industrial audio, where I had no labeled anomalies, no ground truth to lean on, and still reached AUC 0.80 by designing representations that encode only what deviates from normality - exactly the kind of sparse, anomaly-driven signal processing that neuromorphic architectures are built for.

Two further proof points relevant to the Lab's research goals: first, I ran structured **ablation studies** across transformer block unfreezing depth (**4, 6, and 8 layers**), learning rate schedules, and augmentation strategies on a **428M-parameter CLIP ViT-L/14** fine-tuned on **250K samples** - the kind of rigorous experimental methodology required to contribute to peer-reviewed work. Second, at Orange Maroc I engineered a full ML pipeline over **100K+ telemetry records**, applying **unsupervised clustering** to surface latent structure without labeled targets - directly mirroring the unsupervised, data-efficient learning problems central to neuromorphic algorithm research.

Intel's Loihi 2 chip and the Lava software framework represent the most credible open research platform for spiking neural network algorithm development today, and with over 100 peer-reviewed publications behind the program, this is where algorithmic ideas actually get validated at the hardware level. I would welcome the chance to discuss how my background fits the Lab's current research directions.

Sincerely,

AYMANE AIT DADS